

Nipun Attri

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EDUCATION

National Institute Of Technology Karnataka

Bachelor of Technology in Computer Science and Engineering — CGPA : 8.14

Karnataka, Surathkal

Aug. 2025 – Present

Sri Guru Harkrishan Senior Secondary Public School

CBSE, Class XI and XII — Aggregate : 87%

Chandigarh

Aug. 2023 – Mar 2025

EXPERIENCE

Application & Systems Developer

Jan 2026 – Present

IRIS, NITK

- Profiled application infrastructure with **Grafana k6** to identify performance bottlenecks
- Added **monitoring with Prometheus and Grafana** to track system health and performance
- Built a durable **access-log pipeline** using **Vector** and **ClickHouse** for centralized storage and analysis.
- Built **native widgets**, fixed **bugs**, and added **new features** needed by the application

PROJECTS

scx_adaptive: Adaptive CPU Scheduler | C, eBPF, sched_ext, libbpf, Linux | [GitHub](#)

May 2026

- Built an **adaptive, multi-policy CPU scheduler** on **eBPF** with **FCFS**, **Round-Robin**, and **static Priority**
- Wrote a user-space controller that reads **BPF map telemetry** (CPU utilization, burst length, queue latency) and classifies workloads as **CPU-bound**, **interactive**, **batch**, or **stress** using **exponential moving averages**
- Added **hysteresis-gated policy switching** to prevent thrashing on short spikes, plus **SIGINT/SIGTERM handling** that detaches cleanly and returns control to the default kernel scheduler
- Benchmarked with **schbench**: adaptive mode cut **p99 wakeup latency by 55%** (31 μ s vs 69 μ s) and **p99 request latency by 22%** (12.6 ms vs 16.3 ms) compared with FCFS, with **1.5% to 1.8% higher throughput** than all three fixed policies

eBPF XDP and TC Firewall | C, eBPF, XDP, TC, libbpf, Linux | [GitHub](#)

Aug 2026

- Built a **kernel-space packet filter** using **XDP** to drop traffic at the driver level, before SKB allocation
- Implemented **CIDR subnet blacklisting** with a **BPF LPM trie map** and a **CLI** to block, unblock, and list networks at runtime
- Added **DDoS mitigation** with per-source-IP **TCP SYN** and **UDP rate limiting** using atomic counters
- Built **stateful connection tracking** with **TC ingress and egress hooks**, a 5-tuple connection map, and **automatic UDP session timeouts**
- Added **hot configuration reloading** from INI files, so thresholds and blocked protocols update without detaching the programs

ext2 File System Reader and Editor | C++, Linux, File Systems | [GitHub](#)

Nov 2025

- Built a **CLI tool** that parses a raw **ext2 disk image**, reading the **superblock** and **block group descriptors** and printing a **dumpe2fs**-style summary
- Implemented **ls**, **cd**, and **cat** by walking **directory entries and inodes**, including **files that span multiple blocks**
- Added **file write support** that updates data blocks on the disk image

TECHNICAL SKILLS

Languages: C, C++, Dart, Bash

Systems: Linux, eBPF, XDP, sched_ext, libbpf, file systems, operating systems, networking

DevOps & Observability: Docker, Prometheus, Grafana, ClickHouse, Logchef, Vector

Tools: Git, GitHub, Make, clang, gcc, bpftool

Frameworks: Flutter, Ruby on Rails